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Supplemental Information

**Perceptual Decision-Making
as Probabilistic Inference by Neural Sampling**

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717 Supplementary Information

718 S1.1 Sampling equations (related to Experimental procedures)

To simplify notation, we write each image $\mathbf{I}$ as a column vector $\mathbf{y}$, and combine the set of rescaled projective fields $\{\frac{1}{n_x}\mathbf{P}\mathbf{F}_i\}$ into a matrix $\mathbf{G}$. The generative model equations become:

$$D \sim \text{Categorical}(p_D) \quad \text{where } D \text{ is 1 or 2} \quad (\text{S1})$$

$$g_k|D \sim \text{Bernoulli}\left\{\frac{1}{n_g I_0(\kappa)} \exp\left[\kappa \cos 2(\phi_k^{(\mathbf{g})} - \psi_D)\right]\right\} \quad (\text{S2})$$

$$p(x_i|\mathbf{g}) = \frac{1}{\tau_i} \exp(-x_i/\tau_i) H(x_i) \quad (\text{S3})$$

$$\text{with } \tau_i = 1 + \delta \sum_{k=1}^{n_g} g_k \exp\left[\lambda \cos 2(\phi_i^{(\mathbf{x})} - \phi_k^{(\mathbf{g})})\right] \quad (\text{S4})$$

$$p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y} : \mathbf{G}\mathbf{x}, \sigma_y^2) \quad (\text{S5})$$

719 where I_0 is the modified Bessel function of zeroth order, $H(x)$ is the Heaviside function. For
720 simplicity, we assume $\sigma_y^2 = 1$, i.e. the diagonal elements are $\sigma_y = 1$ in the main text.

From this follow the conditional probabilities for Gibbs sampling:

$$p(D|\mathbf{g}) \propto p(\mathbf{g}|D)p_D(D) \quad (\text{S6})$$

$$\propto p_D \prod_{j=1}^{n_g} \begin{cases} \frac{1}{n_g I_0(\kappa)} \exp\left[\kappa \cos 2(\phi_j^{(\mathbf{g})} - \psi_D)\right] & \text{for } g_j = 1 \\ 1 - \frac{1}{n_g I_0(\kappa)} \exp\left[\kappa \cos 2(\phi_j^{(\mathbf{g})} - \psi_D)\right] & \text{for } g_j = 0 \end{cases} \quad (\text{S7})$$

$$p(g_k|D, g_{-k}, \mathbf{x}) \propto p(\mathbf{x}|\mathbf{g})p(g_k|D) \quad (\text{S8})$$

$$= \begin{cases} \frac{1}{n_g I_0(\kappa)} \exp\left[\kappa \cos 2(\phi_j^{(\mathbf{g})} - \psi_D)\right] & \text{for } g_j = 1 \\ 1 - \frac{1}{n_g I_0(\kappa)} \exp\left[\kappa \cos 2(\phi_j^{(\mathbf{g})} - \psi_D)\right] & \text{for } g_j = 0 \end{cases} \quad (\text{S9})$$

$$\times \prod_{j=1}^{n_x} \frac{1}{\tau_j} \exp[-x_j/\tau_j] \quad \text{with } \tau_j \text{ defined as in equation (S4)} \quad (\text{S10})$$

$$p(x_k|\mathbf{g}, x_{-k}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(x_k|\mathbf{g}) \quad (\text{S11})$$

$$\propto \exp\left[-\frac{(\mathbf{y} - \mathbf{G}\mathbf{x})^\top (\mathbf{y} - \mathbf{G}\mathbf{x})}{2\sigma_y^2}\right] \exp\left[-\frac{x_k}{\tau_k}\right] H(x_k) \quad (\text{S12})$$

$$\propto \exp\left[-\frac{\mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{G}\mathbf{x} - \mathbf{x}^\top \mathbf{R}\mathbf{x}}{2\sigma_y^2} - \frac{x_k}{\tau_k}\right] H(x_k) \quad (\text{S13})$$

$$\text{where we have defined: } \mathbf{R} = -\mathbf{G}^\top \mathbf{G} \quad (\text{S14})$$

This means x_k is drawn from a cut-Gaussian distribution (Hoyer and Hyvarinen, 2003) with mean and variance: ^{1 2}

$$\mu = \frac{\mathbf{y}^\top \mathbf{G}_{:,k} + \mathbf{R}_{k,-k} \mathbf{x}_{-k} - \sigma_y^2 \tau^{-1}}{-R_{kk}} \equiv \frac{\sum_i y_i G_{ik} + \sum_{i \neq k} R_{ik} x_i - \sigma_y^2 \tau^{-1}}{-R_{kk}} \quad (\text{S15})$$

$$\sigma^2 = -\frac{\sigma_y^2}{R_{kk}} \quad (\text{S16})$$

We used image patches of size 32x32 pixel, with each dimension scaled from -0.5 to 0.5 (arbitrary units). The PFs consisted of identical 2d-Gabor functions that only differed in their orientation

¹Matching of coefficients of $\frac{-(x-\mu)^2}{2\sigma^2}$ with $Ax^2 + Bx + C$ requires that $\sigma^2 = -\frac{1}{2A}$ and $\mu = \sigma^2 B = -\frac{B}{2A}$. Here, $A = R_{kk}(2\sigma_y^2)^{-1}$ and $B = (2\mathbf{y}^\top \mathbf{G}_{:,k} + \sum_{i \neq k} R_{ik} x_i - 2\sigma_y^2 \tau_k^{-1})(2\sigma_y^2)^{-1}$

²Note: $\mathbf{x}^\top \mathbf{R}\mathbf{x} = \sum_i R_{ii} x_i^2 + \sum_{i \neq j} R_{ij} x_i x_j$

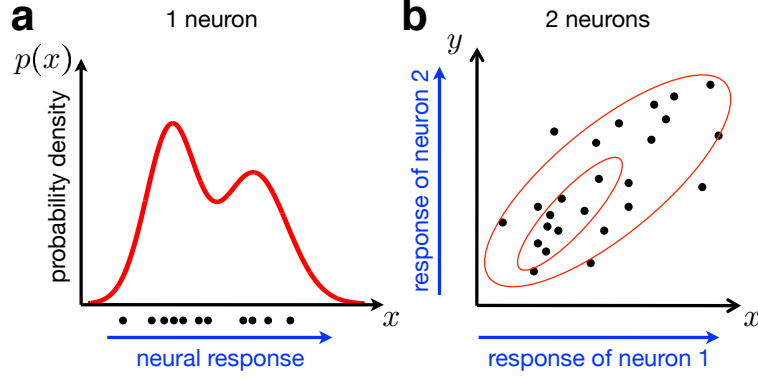


Figure S1: Neural sampling based representation (related to Figures 4-6). **a:** Representation of the posterior over a scalar variable x by the responses of a single neuron. Response variability is related uncertainty in knowledge about x . **b:** If the posterior distribution $p(x, y)$ is correlated, this implies correlations between the neural responses by the two neurons representing x and y , respectively. Note that x and y denote two generic variables here, independent of the model. In our case they may represent (x_i, x_j) for correlations between neural responses, or (x_i, D) to represent correlations between neural responses and behavior.

(equally spaced). Their amplitude was 1, spatial frequency 2, and phase 0. The Gaussian envelope had an standard deviation of 0.1 in the phase-modulated direction, and 0.2 perpendicular to it. Each PF was rescaled such that its 2-norm was 1:

$$\text{PF}_1(x, y) \propto \cos 2x \exp\left(-\frac{x^2}{0.02} - \frac{y^2}{0.08}\right)$$

721 S1.2 Notes on model mismatch and sampling process (related to 722 Results)

In the main text we discussed the sampling process and the reasons for, and effects of, the mismatch between the internal model that the brain performs inference in and the external model defined by the experimenter. The model shown in Figure 2b is easy enough to develop an analytical and conceptual understanding. Since D and g_i are binary variables, we can fully specify this model by:

$$\begin{aligned} p(D = 1) &= p(D = 2) = 0.5 \text{ assuming no bias} \\ p(g_1 = 1|D = 1) &= p(g_1 = 0|D = 2) = a \text{ with } 1/2 \leq a \leq 1 \\ p(g_2 = 1|D = 2) &= p(g_2 = 0|D = 1) = a \\ p(\mathbf{I}|g_1, g_2) &= \mathcal{N}(\mathbf{I} : g_1 \mathbf{PF}_1 + g_2 \mathbf{PF}_2, \mathbf{1}). \end{aligned}$$

From these equations it follows

$$\begin{aligned} p(g_1 = 0|D = 1) &= p(g_1 = 1|D = 2) = 1 - a \text{ and} \\ p(g_2 = 0|D = 2) &= p(g_2 = 1|D = 1) = 1 - a. \end{aligned}$$

723 Here, a parametrizes the mismatch between this model (representing the internal model) and the
724 external, experimenter-defined model. For $a = 1$, both models are identical while values smaller than
725 1 represent an increasing mismatch between the two. Hence the posterior, $p(D, g_1, g_2|\mathbf{I})$ computed

Table 1: Variables and parameters (related to Experimental procedures)

<i>variable</i>	<i>meaning</i>	<i>values</i>
D	decision variable	1 or 2
$\mathbf{g} = (g_1, \dots, g_{n_g})$	grating variables (higher-level visual area)	$g_i = 0$ or 1
$\mathbf{x} = (x_1, \dots, x_{n_x})$	intensity of Gabor elements (V1)	$x_i \in \mathbb{R}^+$
$\mathbf{I}$	stimulus image (retina)	$\mathbf{I} \in \mathbb{R}^N \times \mathbb{R}^N$
$\mathbf{y}$	vector representation of $\mathbf{I}$	$\mathbf{y} \in \mathbb{R}^{N \times N}$
<i>parameters</i>	<i>meaning</i>	<i>values</i>
κ	uncertainty about task-relevant orientations	1
λ	uncertainty about Gabor activation given task-relevant orientation	3
δ	strength of Gabor activation given task-relevant orientation (presence of grating with that orientation)	0.08
$\mathbf{PF}_i$	projective field of neuron i	2-dim Gabor with orientation $\phi_i^{(\mathbf{x})}$
$\mathbf{G}$	matrix with columns representing vectorized projective fields $\mathbf{PF}_i/n_x$	
$\mathbf{R}$	matrix representing the recurrent influences from x_j on x_k	$\mathbf{R} = -\mathbf{G}^\top \mathbf{G}$
n_g	number of possible grating orientations simulated	256 (16 for Figure S2)
n_x	number of Gabor patches (V1 neurons) simulated	1024 (64 for Figure S2)
$N \times N$	size of input image $\mathbf{I}$	32×32
ψ	task-relevant orientations	(45, 135) deg for Figure 4A; (0, 90) deg for Figure 4B; (30, 90, 150) deg for Figure 5A; 90 deg for Figure 5B
$\phi_i^{(\mathbf{g})}$	orientations of gratings	uniformly spaced between 0 and 180 deg
$\phi_i^{(\mathbf{x})}$	orientations of Gabors (= preferred orientations of simulated V1 neurons)	uniformly spaced between 0 and 180 deg
n_s	number of samples drawn per image frame shown	20

by any inference method will be correct with respect to the actual task, and behavioral performance based on it, as measured by % correct, for instance, will be maximal. On the other hand, $a = 0.5$ represents maximal mismatch within the context of this model, and performance will be minimal. In fact, it is easy to see that $p(D|\mathbf{I}) = 0.5$ regardless of the input $\mathbf{I}$ in the case of $a = 0.5$ meaning the model performance will be at chance level. Furthermore, as described in the main text, for $a = 1$, the posterior will only have mass at $(g_1 = 1, g_2 = 0)$ and $(g_1 = 0, g_2 = 1)$ implying a perfect anti correlation between the samples for g_1 and g_2 . Again, at the other extreme for $a = 0.5$, it is easy to see for an ambiguous stimulus $\mathbf{I}$ (e.g. pure noise as used when computing CPs and noise correlations, or equal agreement with $\mathbf{PF}_1$ and $\mathbf{PF}_2$) that $p(g_1, g_2)$ is the same for all values for g_1 and g_2 . This means that the samples for g_1 and g_2 will be uncorrelated. And in both cases – performance and sample correlations – intermediate values $1/2 < a < 1$ will interpolate between the respective extremes.

S1.3 Notes on a sampling-based evidence accumulation (related to Results)

Consider a coarse orientation discrimination task framed in terms of the generative model in Figure 6e. We want to infer the state of a binary orientation variable D given a series of images i.i.d. drawn from a distribution $p(I|D)$. The task is to infer D given a sequence of images:

$$\begin{aligned} p(D|I_1, \dots, I_n) &\propto p(I_1, \dots, I_n|D)p(D) \\ &= p(D) \prod_{i=1}^n p(I_i|D) \end{aligned}$$

Let $p_t(D) \equiv p(D|I_1, \dots, I_t)$, then the following update rule is still exact:

$$\begin{aligned} p_t(D) &\propto p_{t-1}(D)p(I_t|D) \\ &= p_{t-1}(D) \int p(I_t, \mathbf{x}, \mathbf{g}|D) d\mathbf{x} d\mathbf{g} \\ &= p_{t-1}(D) \int p(I_t|D, \mathbf{x}, \mathbf{g})p(\mathbf{x}, \mathbf{g}|D) d\mathbf{x} d\mathbf{g} \\ &= p_{t-1}(D) \int p(I_t|\mathbf{x})p(\mathbf{x}|\mathbf{g})p(\mathbf{g}|D) d\mathbf{x} d\mathbf{g} \end{aligned}$$

In a sampling-based representation, this integral can be computed approximately as

$$\int p(I_t|\mathbf{x})p(\mathbf{x}|\mathbf{g})p(\mathbf{g}|D) d\mathbf{x} d\mathbf{g} \approx \frac{1}{n_s} \sum_{k=1}^{n_s} \frac{p(I_t|\mathbf{x}^{(k)})p(\mathbf{x}^{(k)}|\mathbf{g}^{(k)})p(\mathbf{x}^{(k)}|D)}{p_s(\mathbf{x}^{(k)}, \mathbf{g}^{(k)})}$$

where the samples $(\mathbf{x}^{(k)}, \mathbf{g}^{(k)})$ are drawn from some sampling distribution function $p_s(\mathbf{x}, \mathbf{g})$ (in our case the posterior). We obtain for the log odds based update rule:

$$\log \frac{p_t(D=1)}{p_t(D=2)} \approx \log \frac{p_{t-1}(D=1)}{p_{t-1}(D=2)} + \log \frac{\frac{1}{n_s} \sum_{k=1}^{n_s} \frac{p(I_t|\mathbf{x}^{(k)})p(\mathbf{x}^{(k)}|\mathbf{g}^{(k)})p(\mathbf{x}^{(k)}|D=1)}{p_s(\mathbf{x}^{(k)}, \mathbf{g}^{(k)})}}{\frac{1}{n_s} \sum_{k=1}^{n_s} \frac{p(I_t|\mathbf{x}^{(k)})p(\mathbf{x}^{(k)}|\mathbf{g}^{(k)})p(\mathbf{x}^{(k)}|D=2)}{p_s(\mathbf{x}^{(k)}, \mathbf{g}^{(k)})}}$$

We have assumed that the brain cannot cache multiple (n_s) samples, but that it updates its log odds for the correct decision (the left hand side of the equation) sequentially with every incoming sample. As a consequence, our model approximates the second (update) term on the right hand

side by a sum over n_s terms, each weighted with $1/n_s$:

$$\begin{aligned} \log \frac{p_t(D=1)}{p_t(D=2)} &\approx \log \frac{p_{t-1}(D=1)}{p_{t-1}(D=2)} + \sum_{k=1}^{n_s} \frac{1}{n_s} \log \frac{\frac{p(I_t|\mathbf{x}^{(k)})p(\mathbf{x}^{(k)}|\mathbf{g}^{(k)})p(\mathbf{x}^{(k)}|D=1)}{p_s(\mathbf{x}^{(k)},\mathbf{g}^{(k)})}}{\frac{p(I_t|\mathbf{x}^{(k)})p(\mathbf{x}^{(k)}|\mathbf{g}^{(k)})p(\mathbf{x}^{(k)}|D=2)}{p_s(\mathbf{x}^{(k)},\mathbf{g}^{(k)})}} \\ &\approx \log \frac{p_{t-1}(D=1)}{p_{t-1}(D=2)} + \sum_{k=1}^{n_s} \frac{1}{n_s} \log \frac{p(\mathbf{g}^{(k)}|D=1)}{p(\mathbf{g}^{(k)}|D=2)} \end{aligned}$$

which we can now explicitly write as an update rule that can be applied following each sample k :

$$\log \frac{p_t^{(k)}(D=1)}{p_t^{(k)}(D=2)} \approx \log \frac{p_t^{(k-1)}(D=1)}{p_t^{(k-1)}(D=2)} + \frac{1}{n_s} \log \frac{p(\mathbf{g}^{(k)}|D=1)}{p(\mathbf{g}^{(k)}|D=2)}$$

where $p_t^{(k)}$ as defined in the main text. As a result, the effect of the prior belief on the sensory representation given by $x^{(k)}$ is not compensated for when updating the brain's belief over D leading to a "double-counting" of the prior belief, or confirmation bias.

S1.4 Modeling with input information (related to Experimental Procedures)

We focused on the zero-input information case ("zero-signal") for our main results in Figures 4-7. This is the case most relevant to the data on task-dependent noise correlations (Cohen and Newsome, 2008), CPs and psychophysical kernel (Nienborg and Cumming, 2009; Cohen and Newsome, 2009) to which we compare our model predictions. Here, we demonstrate that our approach for modeling task-dependent influences on sensory processing is robust to non-zero input information, i.e. that we obtain qualitatively correct predictions for stimuli that are informative for the task. We computed the neural population responses for two signal levels (contrast of target Gabor embedded in noise) before and after learning (Figure S2). To account for the varying contrast of the observed image, $\mathbf{y}$, we allow the contrast, $c \in \mathbb{R}^+$, in the generative model, $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y} : c\mathbf{G}\mathbf{x}, \sigma_{\mathbf{y}}^2)$, to vary, and perform inference over it given a particular image $\mathbf{y}$ (Schwartz and Simoncelli, 2001). We assume a temporal prior over c , given by a Gaussian distribution with width 0.1 around a mean given by the contrast in the previous time step.

As expected, we find population responses whose amplitude and variance increases with the signal contrast in the stimulus (Figure S2). Furthermore, the response to task-relevant orientations is higher after learning, compared to with before – all in qualitative agreement with experimental evidence (e.g. (Li et al., 2004)).

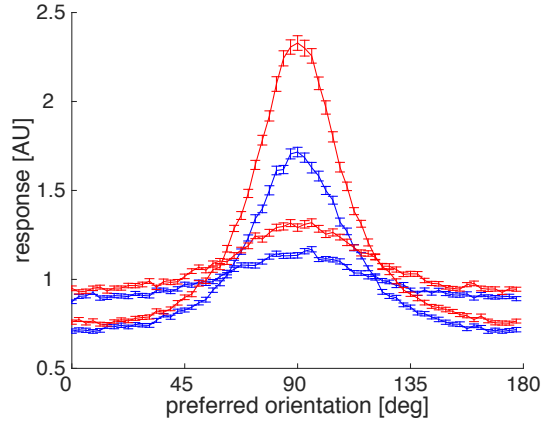


Figure S2: Population response with non-zero signal in the input (related to Figure 8). Blue: Mean responses to two different signal strengths (5 and 10, respectively, in the units of Figure 8D) before learning ($\delta = 0$). Red: Mean responses to the same two contrasts after learning the 0 vs 90 deg orientation discrimination task ($\delta = 0.08$). The error bars indicate standard errors of the mean after 100 repetitions. (The response variability corresponds to error bars 10x that size.)